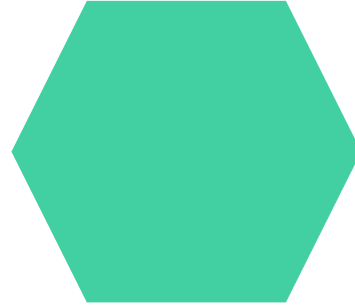




Car Market Trends Analysis

AICTE STU ID :
STU6a6ae9b1894c11785391537

[NITYA NAND TRIPATHI]



PROBLEM STATEMENT



This study aims to analyze the CarDekho used-car dataset to understand the key factors that influence used-car selling prices and market trends. It examines factors such as present price, manufacturing year, fuel type, seller type, transmission, mileage, and ownership to determine how they affect the selling price of used cars.

Understand the dataset and its data quality.

Explore category-wise market and price patterns.

Identify variables strongly associated with selling price.



Project Description

This project performs Exploratory Data Analysis (EDA) on CarDekho used-car records. The analysis includes data loading, inspection, missing-value checking, duplicate removal, IQR-based outlier detection, descriptive statistics and visual analysis.

- 299 records × 9 columns after cleaning
- 0 missing values; 2 duplicate rows removed
- EDA on fuel type, seller type, transmission and selling price

**Workflow: Load → Clean → Inspect → Detect Outliers → Visualize
→ Insights**

WHO ARE THE END USERS?

- Used-car buyers
- Used-car sellers and dealers
- Automobile platforms such as CarDekho
- Market researchers and business analysts
- Data science and machine-learning learners

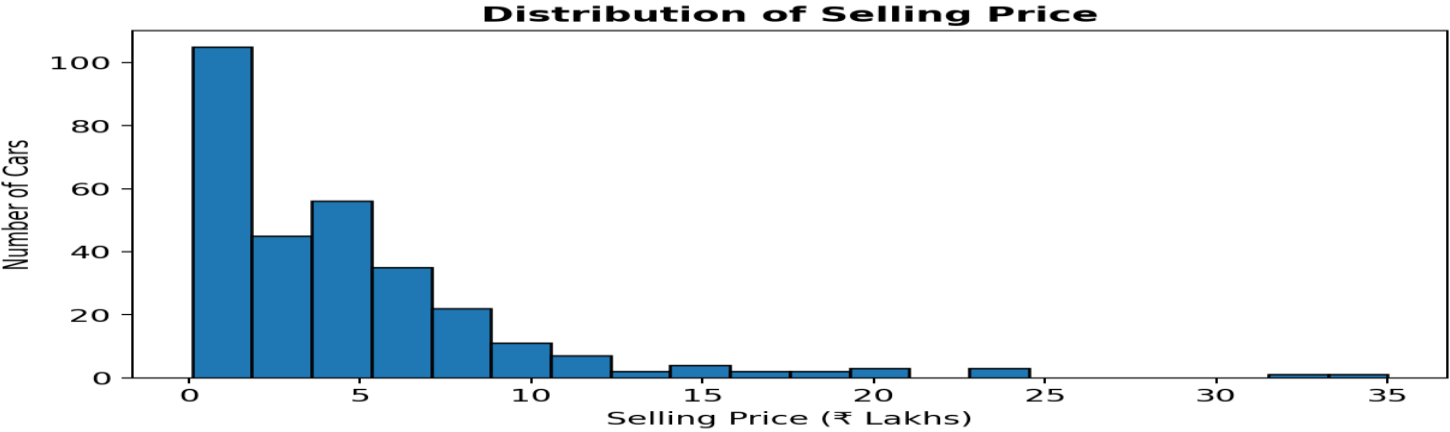
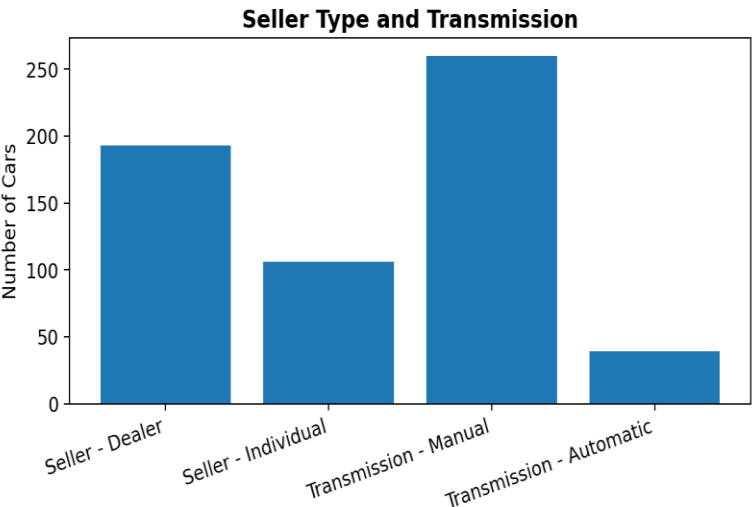
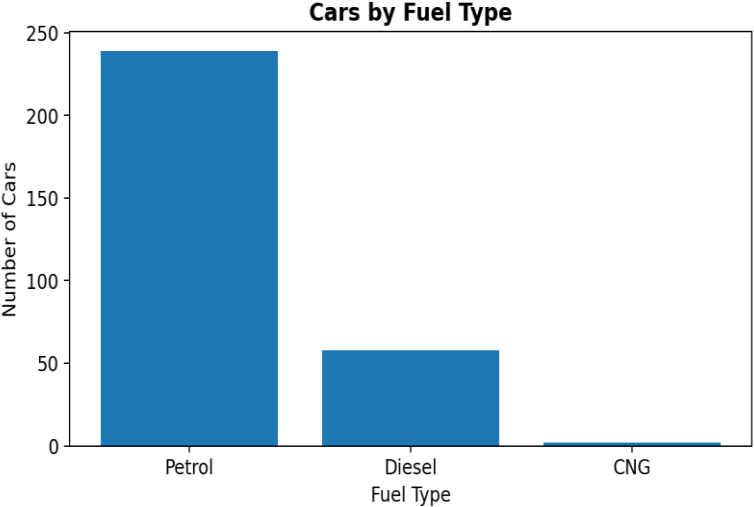
The analysis supports pricing understanding, market segmentation and data-driven decisions.

Technology Used

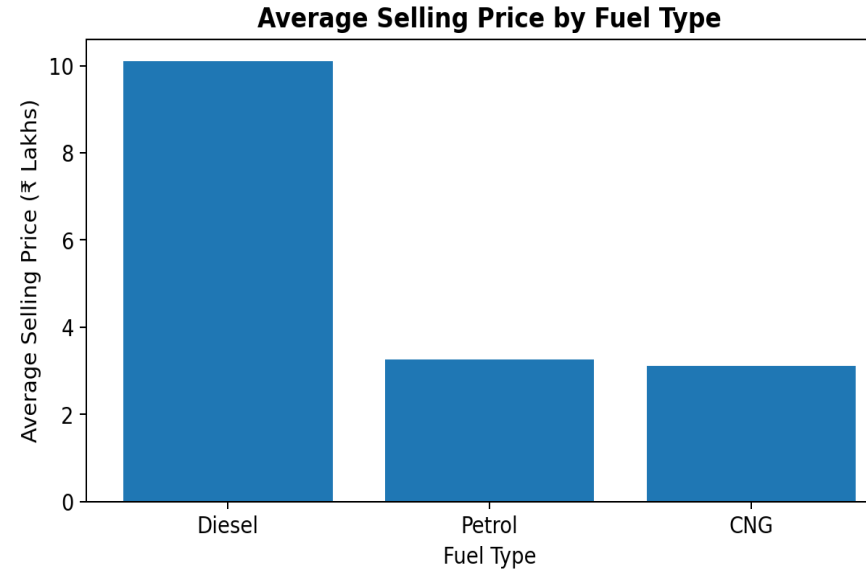
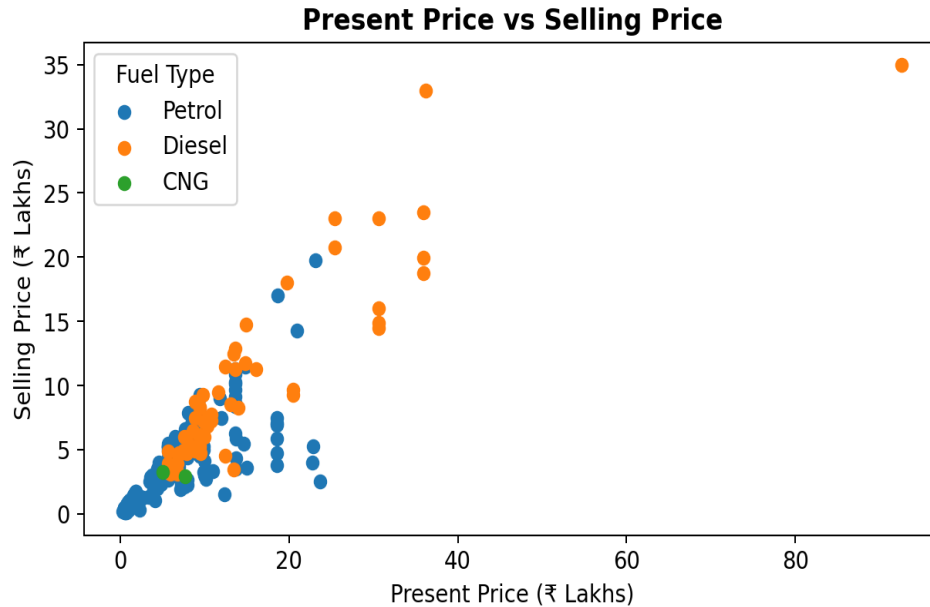
- Python
- Pandas — Data loading and manipulation
- NumPy — Numerical operations
- Matplotlib — Data visualization
- Seaborn — Statistical visualization
- Jupyter Notebook — Analysis environment



RESULTS



RESULTS



- Present Price has the strongest observed relationship with Selling Price ($r \approx 0.88$).
- Diesel cars have the highest average selling price: ₹10.10 lakh.
- Automatic cars average ₹9.07 lakh vs ₹3.92 lakh for manual cars.
- Most frequently listed model: City (26 records).

[Demo Link](#)

Conclusion: Used-car selling prices are most strongly associated with present price, while fuel type and transmission also show clear market-level differences.

Thank you